Q. Define the term Resins?

Q. What are natural resins and synthetic resins? Mention the importance of synthetic resins.

Resins :-

Resins are polymers made by repeatedly discrete molecul (monomers) together to form chain or networks.

Resin are two types :-

(i) Natural resin :-

Natural resin are those that are obtained directly from nature such as fossil remain and tree saps.

Example : Pine sap.

(ii) Synthetic resins :-

Synthetic resin are those that artificially created to mimic the characteristic of its natural occuring counterpart.

Example : Phenol - formaldehyde resin.

Importance of Synthetic resins :-

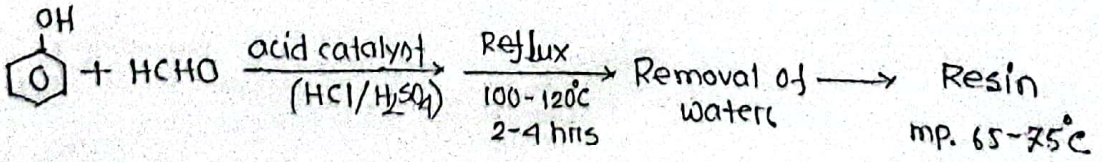
- Synthetic resins are have more uniformity.
- More stable than natural resins.
- They are more pure.
- Economically more favorable.
- Easy to refine.

2018 → Q. Discuss the preparation of phenol-formaldehyde resin novolacs with mechanism.

2015 → Q. Write the preparation of phenol-formaldehyde with mechanism

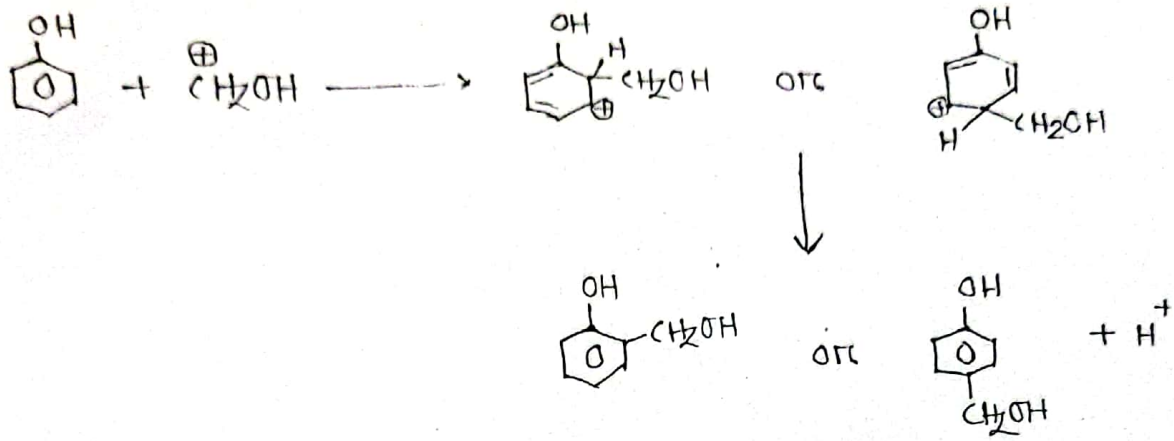
Phenolic resins are prepared by either acid-catalyst (novolacs) or base-catalyzed (resoles) addition of formaldehyde to phenol. by polycondomation reaction.

Acid-catalyzed reaction :-

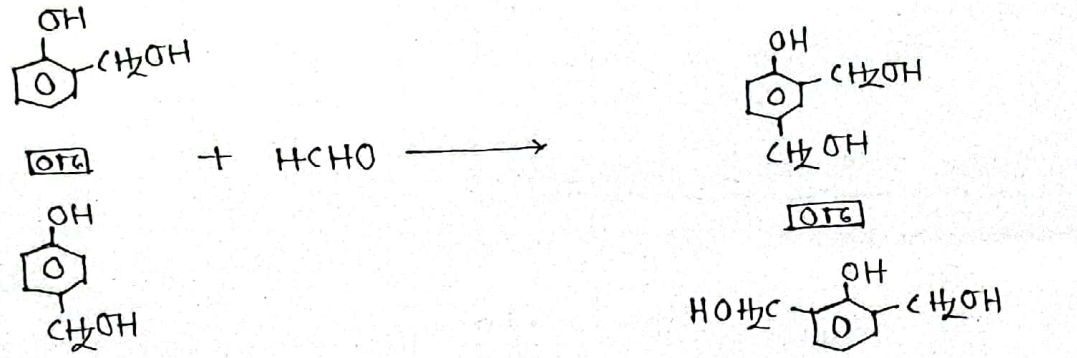


Mechanism :-

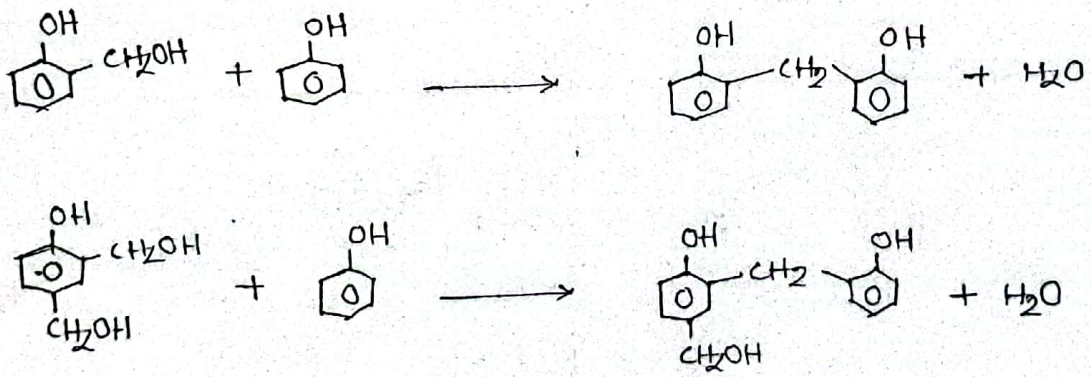
(i) $HCHO + H^+ \longrightarrow {}^{\oplus}CH_2OH$

(ii) 

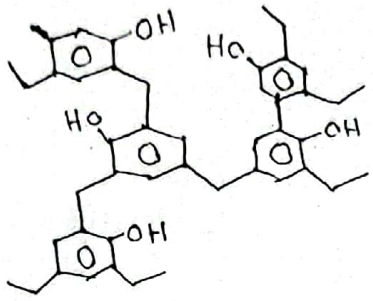

monomethylol phenols.

(iii) 


(Dimethylol phenol)

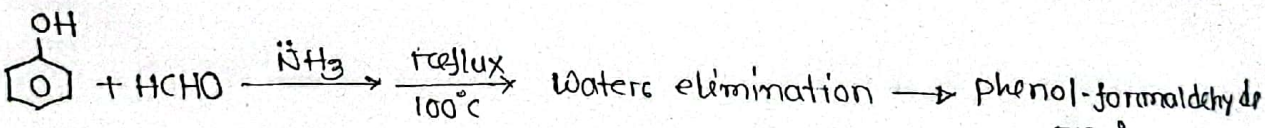
(iv) 

(V) If excess of reagent are added a cross-linked three dimensional polymers will obtain.

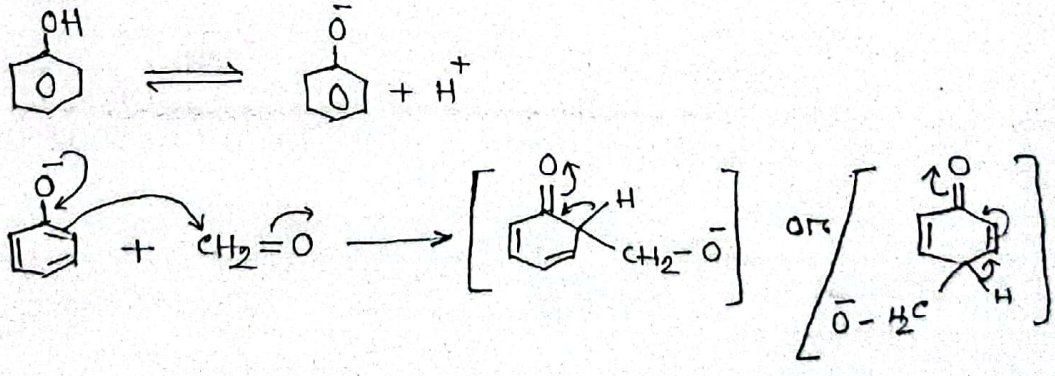


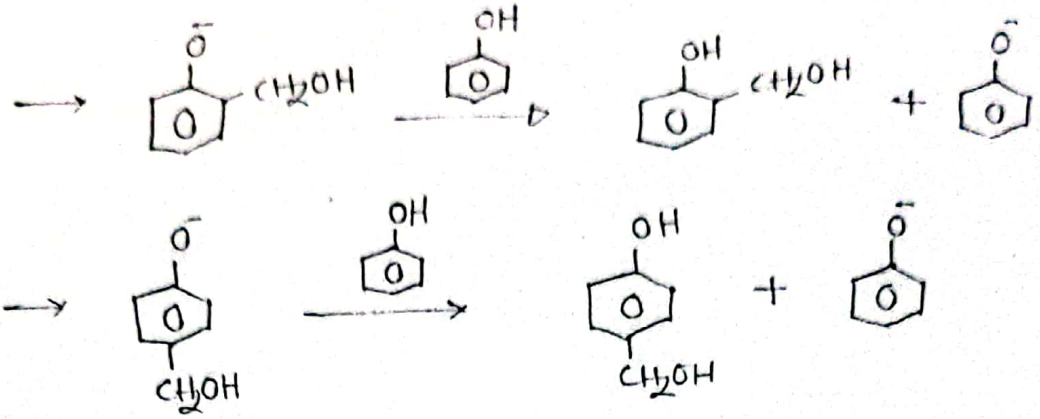
Crosslinked product
(Novlak)

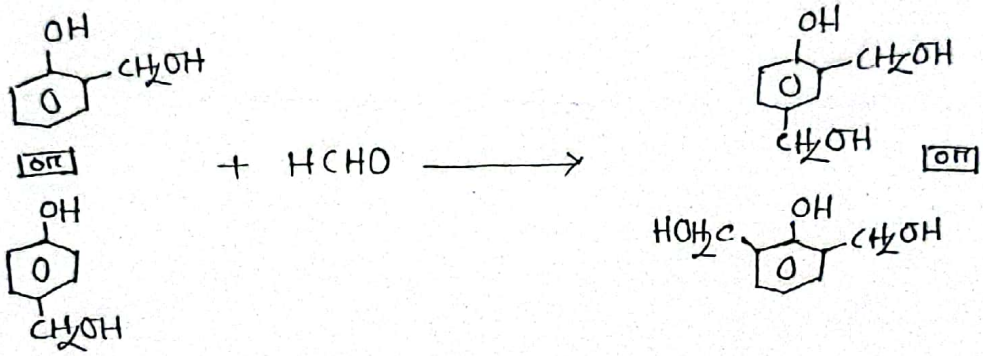
Base catalysed reaction :-



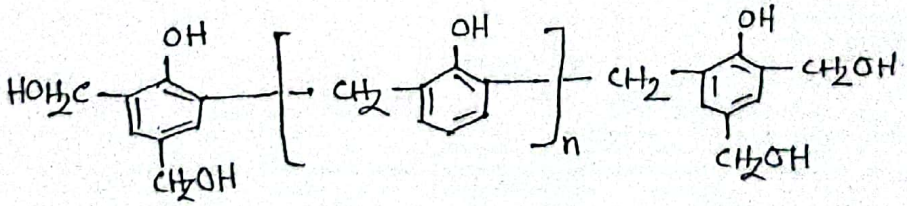

Mechanism :-







The general structure of resol may be as foll



resol.

Melamine-formaldehyde resins :-

Manufacture :-

(a) Preparation of Melamine :-

$$CaC_2 + N_2 \longrightarrow CaNCN \xrightarrow{H_2SO_4} H_2NCN \xrightarrow[100^\circ C]{\bar{O}H} H_2N-C(=NH)-NH-C\equiv N$$

$\xrightarrow{\Delta}$ Melamine

Melamine is also prepare by-

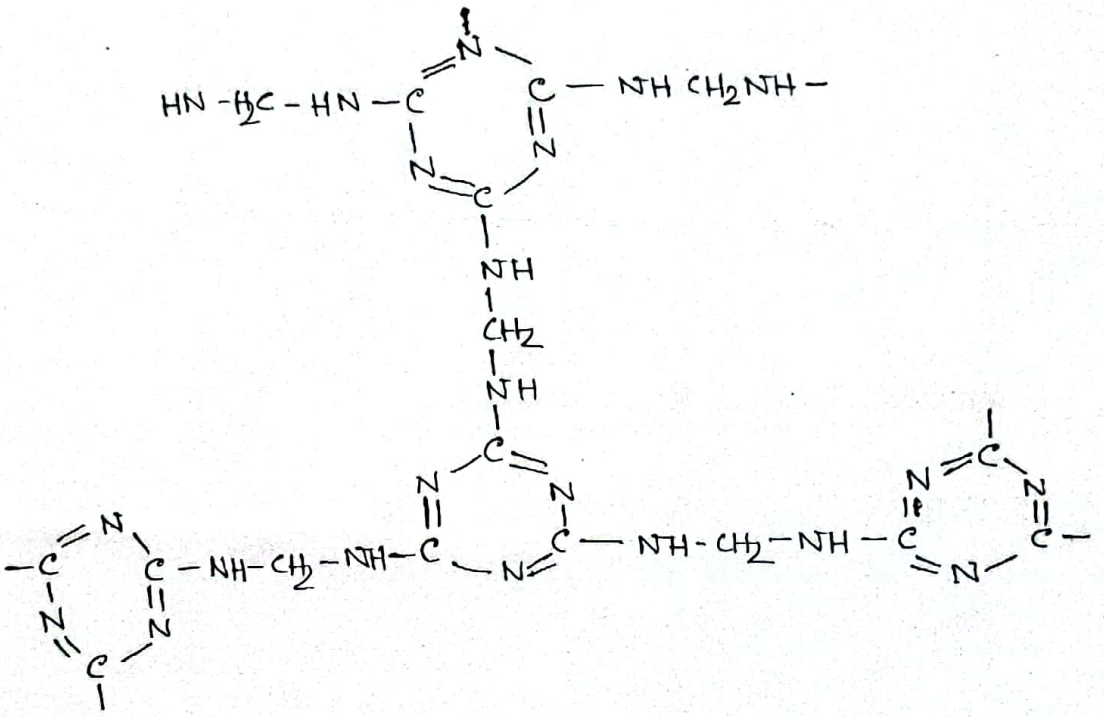
$$2\,NH_2CONH_2 \longrightarrow NH_2CN + NH_2COONH_4$$

$$3\,NH_2CN \longrightarrow$$ Melamine

(b) Preparation of Resin :

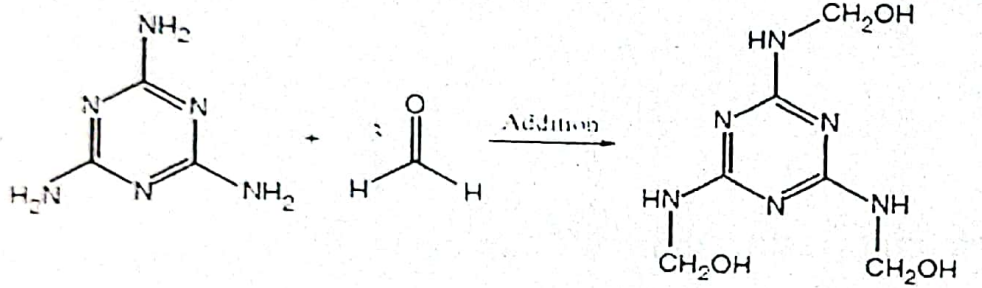
Melamine + $3\,HCHO \xrightarrow[80^\circ C,\ 1\text{-}2\ hrs]{addition}$ Trimethylol melamine

This methylol melamine derivatives reacts with large quan. of melamine will give a linear polymers. This polymers w ~~further~~ react with additional quantities of formaldeh. will give a three dimentional network.

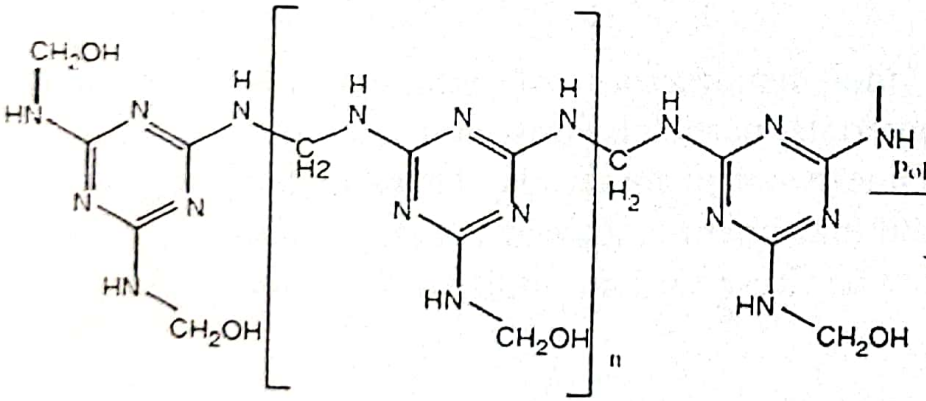


# Melamine-formaldehyde resins

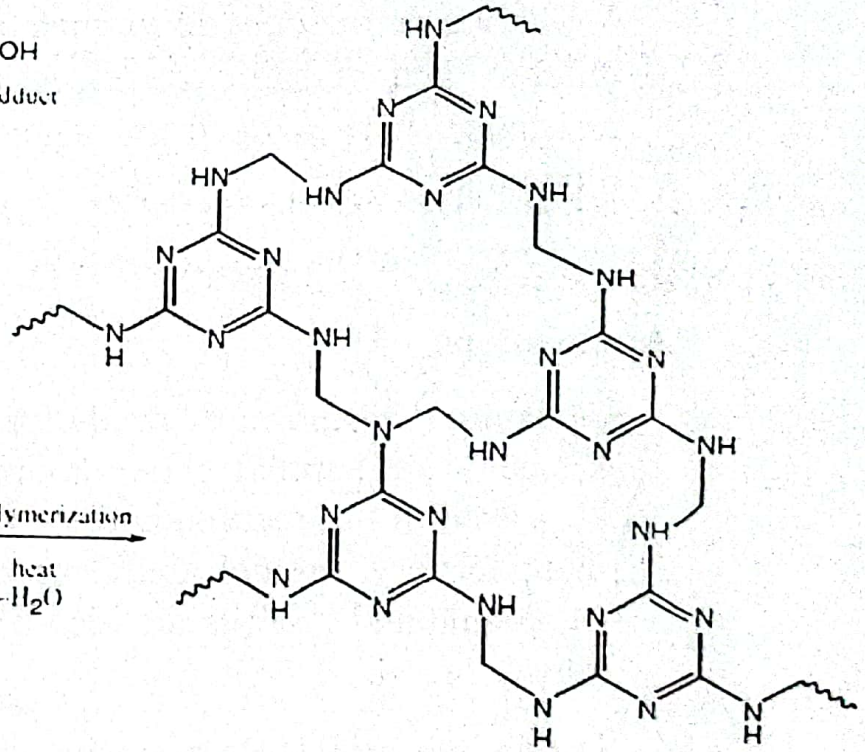
Production of melamine–formaldehyde polymers involves reactions of initial production of methylol derivatives of melamine, which on subsequent condensation, ultimately form methylene bridges between melamine groups in a rigid network structure.




Melamine Formaldehyde Adduct




Melamine Formaldehyde Oligomer




Melamine Formaldehyde Crosslinked Polymer

## Properties

Melamine formaldehyde or melamine resin (short Melamine or MF) is a hard, very durable, and versatile thermosetting plastic (aminoplast[1]) with good fire and heat resistance. It is made from melamine and formaldehyde by condensation of the two monomers.

## Applications

Melamine resins are used for the manufacture of many products, including kitchenware, laminates, overlay materials, particleboards, and floor tiles. Melamine is also used in the manufacture of flame-resistant materials. These include textiles such as upholstery, firemen uniforms, thermal liners, and heat resistant gloves and aprons. Melamine and its salts are also used as fire-retardant additives in paints, plastics, and paper.

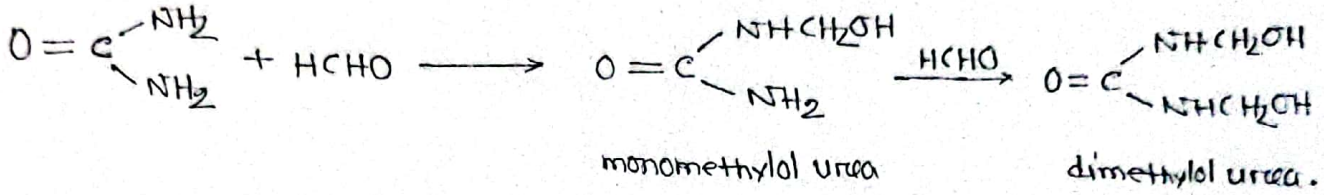
Melamine-formaldehyde resins are based on melamine and formaldehyde Noted for their excellent water resistance, Used for dishwater safe dinner ware which can be decorated with molded-in paper overlays, Form the surface layer for decorative laminates (Formica) Used as an adhesives for water resistant plywood.

## Urea- formaldehyde resin:-

2014 → 4.
Q. Write the preparation of urea formaldehyde resin with mechani

2014 → 2.
Q. Discuss the preparation and uses of urea formaldehyde resin.

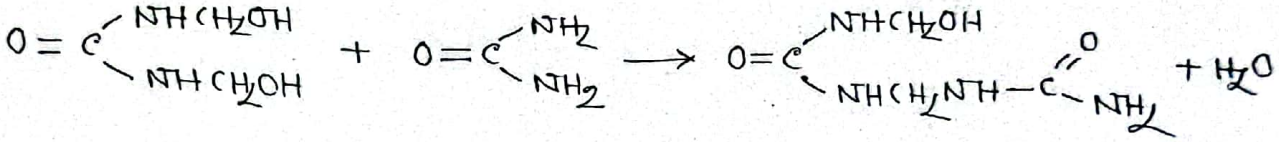
**Step-1:**



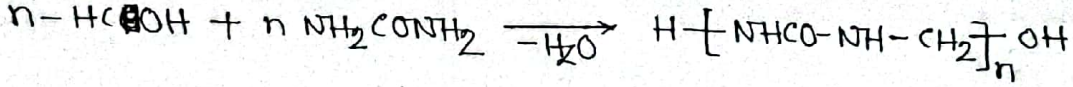

monomethylol urea

dimethylol urea.

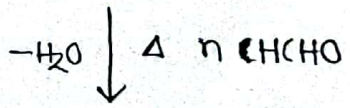
**Step-2:**

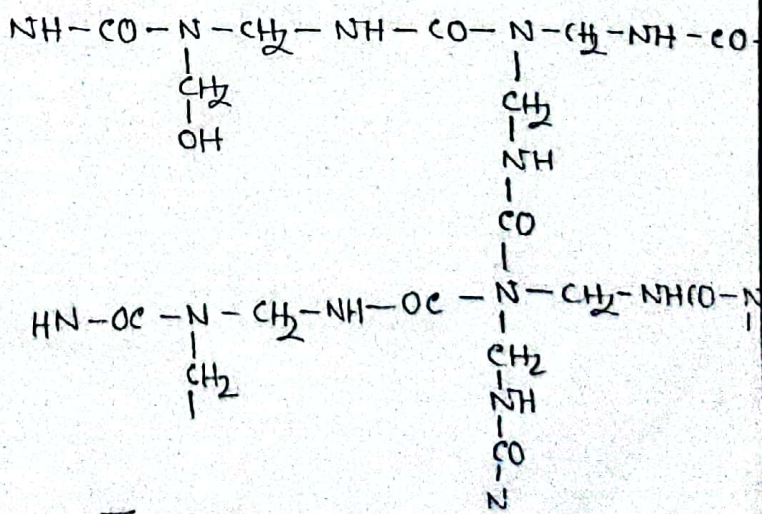



**Step-3:-**




Linear structure



Three dimentional network.

## PROPERTIES OF UREA FORMALDEHYDE RESIN

- It has a very high tensile strength.
- Urea formaldehyde has the property of flexural modulus.
- Has the property of heat distortion temperature.
- Has the capacity of low water absorption.
- It has the property of mould shrinkage.
- Has a property of high surface hardness.
- It can be elongated at break
- It is volume resistance in nature.

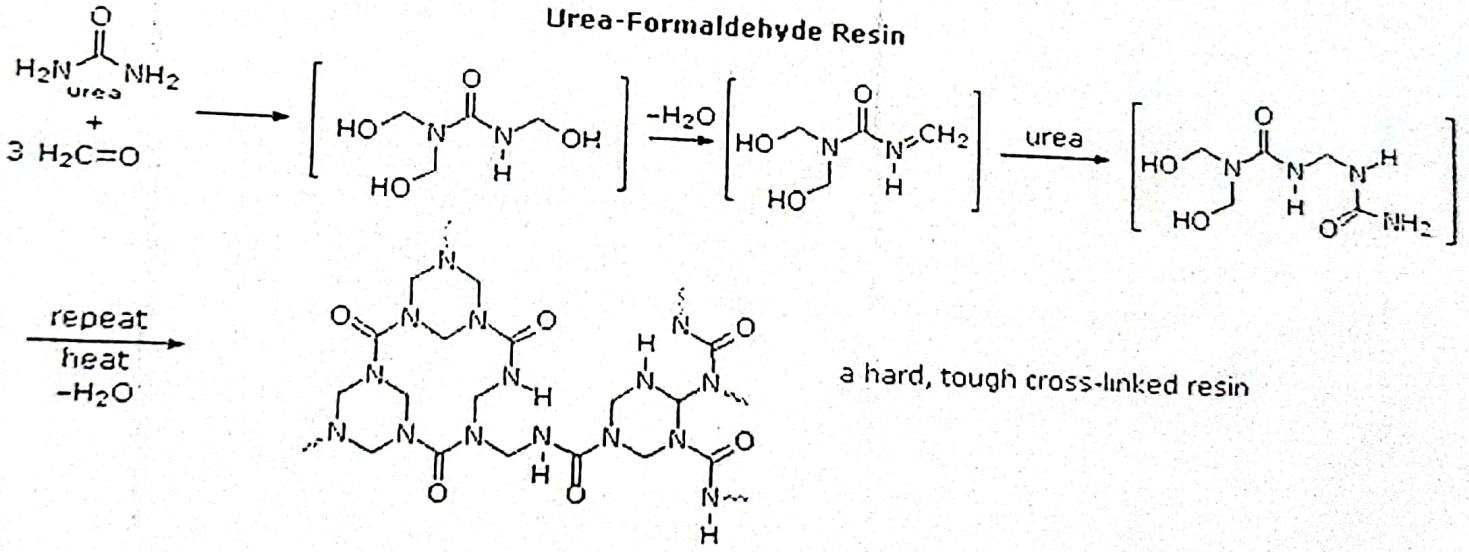
## USES OF UREA FORMALDEHYDE RESIN

Urea formaldehyde is the very common chemical and is mostly used because of its chemical properties. Examples are textiles, paper, foundry sand molds, wrinkle resistant fabrics, cotton blends, rayon, corduroy, etc. also used to glue wood together. Urea formaldehyde is mostly used when producing electrical appliances casing also desk lamps .

Urea formaldehyde is also used in agricultural field as a source of nitrogen fertilizer and Its rate of decomposition is into $CO_2$ and $NH_3$ and is determined by the action of microbes found naturally in soils. The activity of these microbes, and the rate of nitrogen release is temperature dependent. The optimum temperature for microbe activity is approximately about (70-90) °F.

# Urea–Formaldehyde Resins

Urea–formaldehyde resin synthesis consists basically of two steps. In the first step, urea reacts with aqueous formaldehyde under slightly alkaline conditions to produce methylol derivatives of urea In the second step, condensation reactions between the methylol groups occur under acidic conditions, leading ultimately to the formation of a network structure

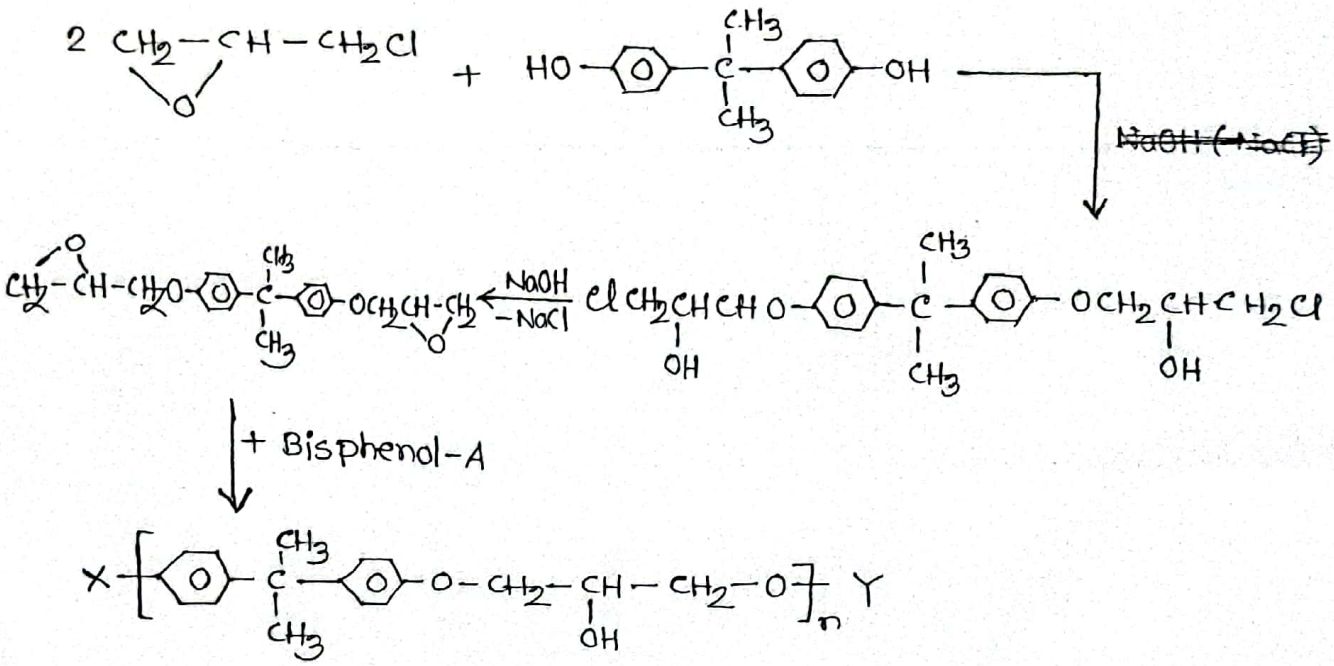


Epoxy resin :-

2015 Q. Discuss the preparation of epoxy resin with mechanism. → 5

2015 Q. Mention few properties and use of epoxy-resins. → 2

Epoxy resins are basically polyethers. It is prepared from epichlorohydrin and bisphenol-A.




Where, X = $CH_2-CH-CH_2-O$ (O) and Y =

## Properties

**High adhesive strength** and high **mechanical properties** are also enhanced by high **electrical insulation** and **good chemical resistance** and **heat resistance**.

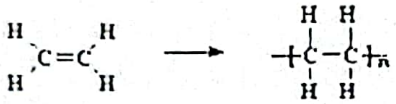
## Applications

Epoxies find uses as adhesives, caulking compounds, casting compounds, sealants, varnishes and paints, as well as laminating resins for a variety of industrial applications.

1. ~~$CH_2-\dot{C}H_2 + \sim CH_2-CH_2-CH_2-CH_2-R$

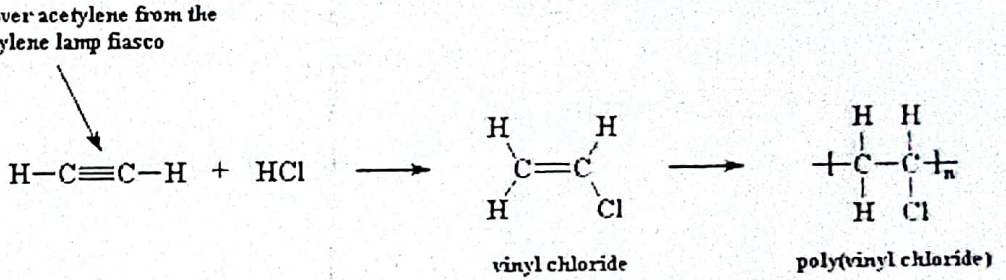
1 -

## Polyethylene

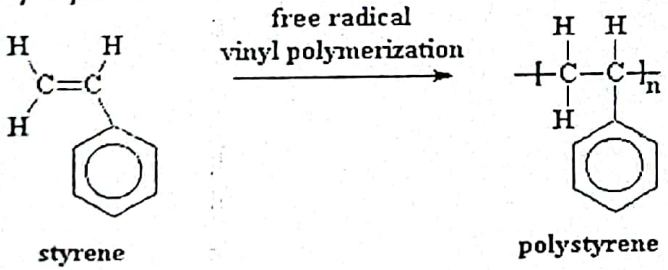


## PVC

leftover acetylene from the acetylene lamp fiasco

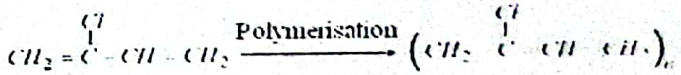



## Polystyrene

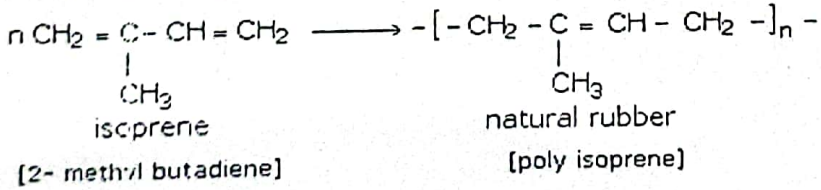



## Synthetic rubber

Chloroprene undergoing trans-polymerization to produce synthetic rubber, polychloroprene




## Natural rubber




Polyester, Polyamide

## Polyethylene

Polyethylene is the simplest hydrocarbon polymers and have the following structure:-

$$-\left[CH_2-CH_2\right]_n-$$

Two types of polyethylenes-

(i) low density polyethylene (LDPE) → branching

(ii) High density polyethylene (HDPE). → Linear.

G-216 **Low density polyethylene:-**

Low density polyethylene is produced by the high pressure polymerization (~1500 atm) at 110-125°C temp.

$$\text{Ethylene} \xrightarrow[110-125^\circ C]{\text{At high pressure}} \text{LDEP.}$$

The branching occurs by intramolecular or intermolecular chain transfer reactions as follows:-

1. $\sim\sim CH_2-\dot{C}H_2 + \sim CH_2-CH_2-CH_2-CH_2-R$

↓ Intermolecular chain transfer

$\sim\sim CH_2-CH_3 + \sim\sim CH_2-\dot{C}H-CH_2-CH_2-R.$

2.

$\sim CH_2-CH_2-CH_2-CH_2-CH_2-\dot{C}H_2$

↓ Intramolecular chain transfer

$\sim CH_2-CH_2-CH_2-\dot{C}H-CH_2-CH_3$.

Mechanism:-

Initiation:-

$I-I \longrightarrow \dot{I}$

$\dot{I} + CH_2=CH_2 \longrightarrow I-CH_2-\dot{C}H_2$

Propagation:

$I-CH_2-\dot{C}H_2 + CH_2=CH_2 \longrightarrow I-CH_2CH_2CH_2\dot{C}H_2$

. . . . . .

. . . . . .

$I-(CH_2-CH_2)_{n-1}-\dot{C}H_2 + CH_2=CH_2 \longrightarrow I-(CH_2-CH_2)_n-\dot{C}H_2$

Termination:-

$\sim CH_2-\dot{C}H_2 + \dot{C}H_2-CH_2\sim \longrightarrow \sim CH_2 + CH_2-CH_2\sim$

$\sim CH_2-\dot{C}H_2 + \dot{C}H_2-CH_2\sim \longrightarrow \sim CH_2-CH=CH_2 + CH_2=CH\sim$

Properties:-

(i) good toughness

(ii) good electrical properties

(iii) translucent due to crystallinity

Uses :-

(i) packing and wrapping frozen food

(ii) textile products.

(iii) Pipes used for both agricultural and domestic water line

(iv) Insulation to electrical cables.

poly(vinyl chloride) :-

2015 → 2 Q. Discuss the properties and uses of poly(vinyl chloride).

Vinyl chloride is obtained from -

(i) $CH \equiv CH + HCl \xrightarrow[130^\circ C,\ 5atm]{Hg_2Cl_2} CH_2=CHCl$

(ii) $CH_2=CH_2 + Cl_2 \longrightarrow CH_2Cl-CH_2Cl \longrightarrow CH_2=CHCl + HCl.$

$$CH_2=CHCl \xrightarrow{\text{polymers}} \sim CH_2-\underset{Cl}{CH}-CH_2-\underset{Cl}{CH}\sim$$

poly(vinyl chloride).

Vinyl chloride
+
trichloromethylene
+
Lauryl peroxide
→ Reactors → PVC colorless.

Properties :-

(i) Colorless rigid material
(ii) High density
(iii) low softening point
(iv) chemically inert.

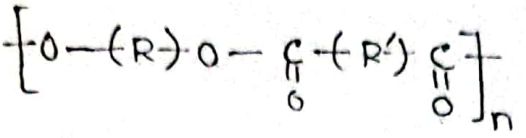
Uses :-

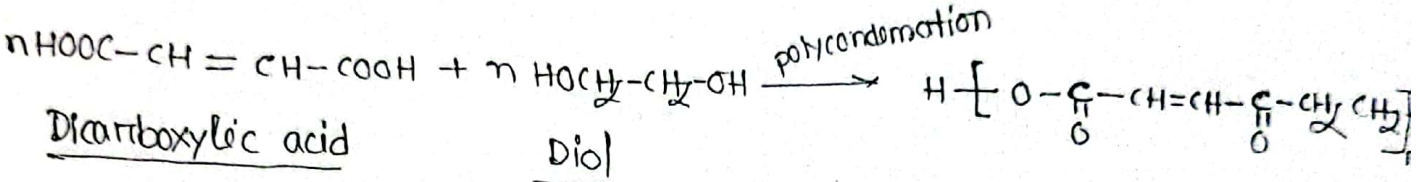
(i) Cable. insulation
(ii) Leather cloth
(iii) packaging and toys.
(iv) making rain wear
(v) hand bages
(vi) ~~Sue~~ food covers etc.

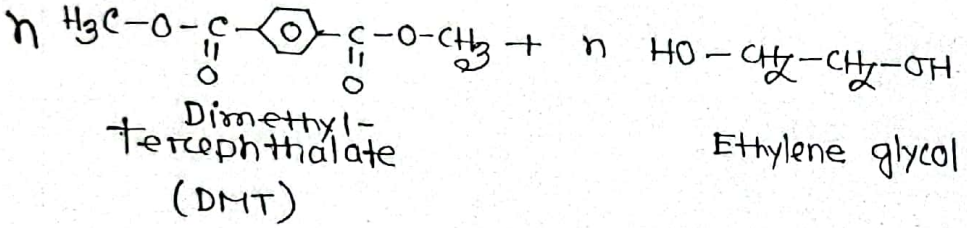
PTFO-
317.

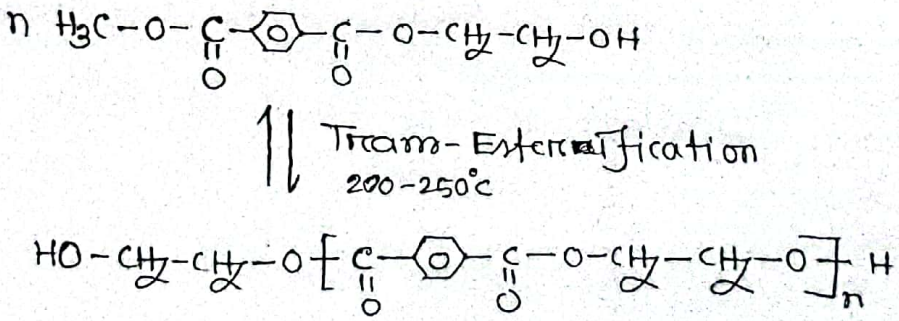
Polyesters :-

Polyesters have the structure with ester linkages as follows:






Polyethylene terephthalate (PETP).

Properties :-

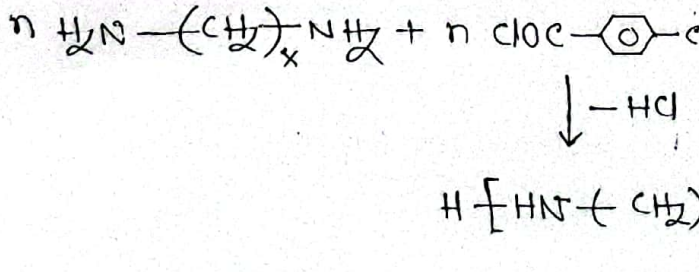
① Resistance to heat and moisture
(ii) remain virtually unattacked by most of the chemicals.
(iii) good electrical and thermal properties

Uses :-

(i) Used to make textile fibers.
(ii) magnetic recording tapes
(iii) typewriter ribbon
(iv) food packing
(v) book covers etc.

Polyamide :-

Polyamide can be obtained by the melt polycondensation between dicarboxylic acid and diamine.



Properties :-

(i) high impact strength
(ii) toughness
(iii) flexibility
(iv) abrasion resistance.
(v) High melting point
(vi) High tensile strength

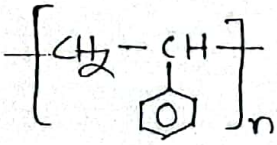
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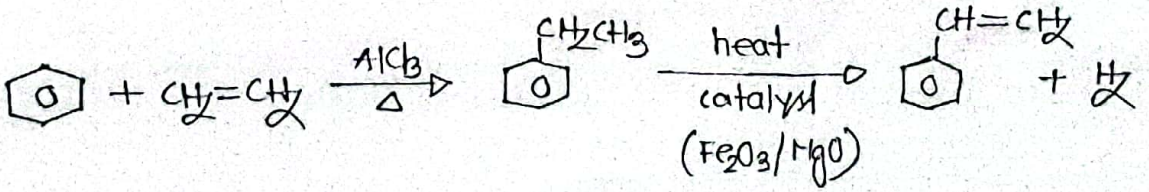
Uses :-

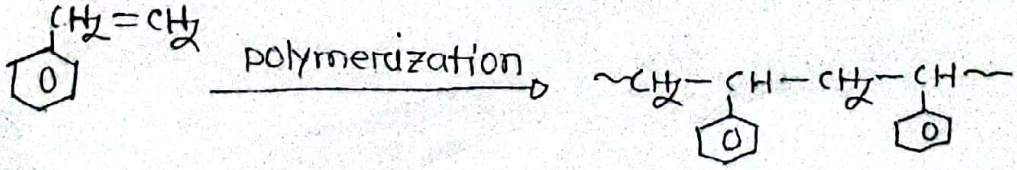
(i) Used in gears, cams, brushes
(ii) used in hypodermic syringes and surgical accessories.

Polystyrene :-

Polystyrene is also known as poly(vinyl benzene). as follow





~~Application :-~~

Properties :-

(i) Hard, rigid, and brittle

(ii) Highly transparent to visible light

(iii) good insulator

(iv) low moisture absorption

Uses :-

(i) Making household goods

(ii) Lenses

(iii) plastic

(iv) toys

(v) wood laminates

(vi) films

(vii) Battery boxes

(viii) Electrical components.

অসমাপ্ত
30.01.20 2:30 am